

STRING MATCHING APPLICATION

Team Members:

J. SIMRAN - 2520030166

K. ANUSRI - 2520030432

SIRIPURAPU RIKHIL - 2520030480

ABSTRACT

Our project aims to develop a document similarity checking application using the Z-Algorithm. The application compares the textual content of two PDF documents to identify matching or duplicated content. By extracting text from the uploaded PDFs and applying efficient string-matching techniques, the system can determine the similarity between the documents and display the matching portions.

PROBLEM STATEMENT

In a college, an assignment is given to students with a requirement to submit an individual project report. Student A completes the assignment genuinely by doing their own work and prepares a PDF report. Student B does not complete the work independently and asks Student A for their PDF. Student B then copies the entire document and only changes personal details such as the name and roll number before submitting it as their own work.

When the instructor receives both submissions, manually comparing lengthy PDF documents to identify such copied content is time-consuming and difficult.

Therefore, an efficient system is needed to compare two PDF documents, identify the matching content, and calculate their similarity percentage.

PROPOSED FUNCTIONALITIES

1. PDF Upload

Upload two PDF documents for comparison.

2. Text Extraction

Extract textual content from both PDFs.

3. Z-Algorithm Based Matching

Compare the extracted text using the Z-Algorithm.

4. Similarity Calculation

Calculate the percentage of matching content.

5. Visual Results

Display the similarity percentage and matching portions.

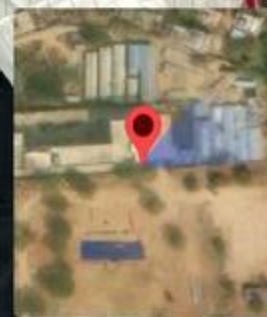
TOOLS AND TECHNOLOGIES

TOOLS	TECHNOLOGIES
Z-Algorithm	Core string-matching algorithm
PDF Text Extraction	Extract text from uploaded PDFs
Document Comparison	Compare extracted text and identify matching content
User Interface	Upload PDFs and display similarity results

EXPECTED OUTCOME

- **Compare** two PDF documents and identify similar content.
- **Generate** a similarity percentage based on the matching text.
- **Identify** and display matching portions of the documents.
- **Efficiently** detect copied or duplicated content using Z-Algorithm-based string matching.

Expected Result: A simple and efficient document similarity-checking application.



Hyderabad, Telangana, India 🇮🇳

Dollar Dreams, Bowrampet, Hyderabad, Telangana 500043, India

Lat 17.547206, Long 78.404749

Thursday, 20/08/2026 01:30 PM GMT+05:30

Note : Captured by GPS Map Camera



GPS Map Camera

THANK YOU!